

Boppudi Omsaichand

✉ saichandboppudi@gmail.com — 📞 +91-9346859956
Andhra Pradesh, India — 🌐 github.com/omsaichand35

Professional Summary

Computer Science undergraduate passionate about Artificial Intelligence, Machine Learning, Computer Vision and Agentic AI systems. Experienced in developing production-ready mobile applications, Retrieval-Augmented Generation (RAG) pipelines, database agents, reinforcement learning systems, and multimodal AI applications. Interested in building scalable AI products that combine LLMs, computer vision, and intelligent automation.

Education

Indian Institute of Information Technology Dharwad

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 7.77

Intermediate (State Board): 97.9%

CBSE (Class X): 100%

Technical Skills

Languages

Python, Java, Kotlin, C++, C, SQL

AI / ML

PyTorch, Scikit-learn, Deep Learning, Reinforcement Learning, OpenCV, MediaPipe, Image Processing

LLM / GenAI

LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, MCP (Model Context Protocol), Ollama

Frameworks

Android (Kotlin), Room Database, SQL Server

Developer Tools

Git, CUDA, Linux, WinForms

Experience

Mobile Application Developer

Warehouse Tracking System

- Developed an Android-based warehouse management application using Kotlin and SQL Server.
- Implemented barcode-based inventory management for production warehouses.
- Built real-time stock tracking and automated inventory workflows.
- Integrated geolocation verification to ensure worker presence during warehouse operations.
- Reduced manual operational effort by approximately 60–70%.

Projects

Multimodal Retrieval-Augmented Generation (RAG) System

- Developed a local multimodal RAG pipeline capable of understanding PDFs, images, text documents and structured files.
- Implemented document ingestion, OCR, embedding generation, semantic search and context-aware response generation.
- Designed retrieval pipelines supporting multiple input modalities with efficient local inference.
- Integrated lightweight LLMs for offline document question answering.

Natural Language to SQL Agent using MCP

- Built an MCP-powered AI agent that converts natural language questions into executable SQL queries.
- Implemented schema-aware prompt generation using automatic database introspection.
- Added SQL validation, execution, result explanation and secure query workflows.
- Designed extensible MCP tools supporting multiple databases and local LLM backends.

Autonomous Navigation using Deep Reinforcement Learning

- Developed a reinforcement learning navigation agent for dynamic environments.
- Optimized reward engineering and environment design to improve policy convergence.
- Focused on reducing training time while improving policy stability.

Warehouse Management System (Production Deployment)

- Built and deployed an Android warehouse application used in production environments.
- Implemented barcode scanning, inventory synchronization and SQL Server integration.
- Automated warehouse workflows with real-time inventory updates.

Semantic Segmentation of Satellite Imagery

- Evaluated multiple pre-trained segmentation networks on satellite datasets.
- Compared models using IoU and pixel-level accuracy.
- Investigated applications in remote sensing and land-use analysis.

Agricultural Land Abandonment Prediction

- Performed time-series analysis on agricultural datasets.
- Applied machine learning methods for trend prediction and pattern analysis.

Achievements

- Secured **4th Place** in a National Hackathon.
- Achieved **1st Place** in an individual problem statement competition.
- Invited to conduct research on optimizing Deep Reinforcement Learning through reward engineering and environment optimization.
- Successfully deployed a warehouse management application reducing manual labor by approximately 60–70%.

Areas of Interest

Artificial Intelligence
Machine Learning
Generative AI
Agentic AI
Computer Vision
Large Language Models